Edge-Generated Localization and Identification of Road Potholes via Real-Time Vehicle Dashcam Footage for Road Maintenance and Safety

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Potholes represent a persistent threat to both the fiscal stability and physical safety of communities across the United States, costing American motorists approximately $26.5 billion in 2021 with individual repair costs averaging nearly $600 per incident. In order to mitigate this infrastructure challenge, an automated, AI-driven pothole detection and geospatial mapping framework, Smooth Cruize, is proposed. In this study, it is hypothesized that leveraging real-time video analytics via the YOLO (You Only Look Once) architecture and evaluating hazard severity through Gemini AI will provide a highly precise, publicly accessible interface for autonomously cataloging road degradation. The framework utilizes integrated GPS data to map hazards and serves as an alternative to traditional, manual road hazard reporting. Preliminary empirical testing demonstrated that the integration of YOLO and Gemini AI provides a high degree of precision in identifying and evaluating infrastructure hazards. The main conclusion of this project is that automating hazard detection through artificial intelligence offers a highly efficient way to empower communities and assist government agencies, such as the Department of Transportation, in prioritizing infrastructure repairs. Future research and development will focus on the deployment of a mobile application designed to provide proactive alerts to motorists, alongside exploring the integration of this technology into public transit infrastructure.

Sources: American Automobile Association (AAA) Pothole Damage Statistics: https://newsroom.aaa.com/2022/03/aaa-potholes-pack-a-punch-as-drivers-pay-26-5-billion-in-related-vehicle-repairs/

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